

PACK INSERT FOR MALAYSIA

VALCYTE®

Valganciclovir

1. DESCRIPTION

1.1 Therapeutic/Pharmacologic Class of Drug

Antiviral

ATC code: J05AB14

1.2 Type of Dosage Form

Film-coated tablet

1.3 Route of Administration

Oral

1.4 Qualitative and Quantitative Composition

Active ingredient: valganciclovir hydrochloride

Film-coated tablets: 450 mg

The tablets are pink, film-coated, and convex oval with ‘VGC’ embossed on one side and ‘450’ on the other side.

Excipients:

Tablet:

Tablet core: povidone, crospovidone, microcrystalline cellulose, stearic acid powder

Tablet coat: hydroxypropyl methylcellulose, titanium dioxide (E171), polyethylene glycol, red iron oxide (E172), polysorbate

2. CLINICAL PARTICULARS

2.1 Therapeutic Indication(s)

ValcYTE is indicated for the treatment of cytomegalovirus (CMV) retinitis in acquired immunodeficiency syndrome (AIDS) patients.

ValcYTE is indicated for the prevention of CMV disease in CMV-negative patients who have received a solid organ transplant from a CMV-positive donor.

2.2 Dosage and Administration

Cautions – Strict adherence to dosage recommendations is essential to avoid overdose.

Standard Dosage

ValcYTE is administered orally, and should be taken with food (see Sections 3.2.5, *Pharmacokinetics in Special Populations*, and 3.2.1, *Absorption*).

ValcYTE is rapidly and extensively converted into the active ingredient ganciclovir. The bioavailability of ganciclovir from ValcYTE is up to 10-fold higher than from oral ganciclovir.

The dosage and administration of ValcYTE tablets as described below should be closely followed (see Sections 2.4, *Warnings and Precautions*, and 2.7, *Overdose*).

Treatment of cytomegalovirus (CMV) retinitis

Induction treatment of CMV retinitis:

For patients with active CMV retinitis, the recommended dose is 900 mg twice a day for 21 days. Prolonged induction treatment may increase the risk of bone marrow toxicity (see Section 2.4, *Warnings and Precautions*).

Maintenance treatment of CMV retinitis:

Following induction treatment, or in patients with inactive CMV retinitis, the recommended dose is 900 mg once daily. Patients whose retinitis worsens may repeat induction treatment (see *Induction treatment of CMV retinitis*).

The duration of maintenance treatment should be determined on an individual basis.

Prevention of CMV disease in solid organ transplantation: For kidney transplant patients, the recommended dose is 900 mg once daily starting within 10 days of transplantation until 200 days post-transplantation.

For patients who have received a solid organ transplant other than kidney, the recommended dose is 900 mg once daily starting within 10 days of transplantation until 100 days post-transplantation.

2.2.1 Special Dosage Instructions

Geriatric Use

Age and efficacy have not been established in this patient population. No studies have been conducted in adults older than 65 years of age. Since renal clearance decreases with age, ValcYTE should be administered to elderly patients with special consideration of their renal status (see Table 1 and section 3.2.5 *Pharmacokinetics in special populations, Geriatric Population*).

Adult Patients with renal impairment

Serum creatinine or creatinine clearance levels should be monitored carefully. Dosage adjustment is required for adult patients based on creatinine clearance as shown in the table below (see Sections 3.2.5, *Pharmacokinetics in Special Populations* and 2.4, *Warnings and Precautions*).

Table 1: ValcYTE Tablets Dose for Renally Impaired Patients

CrCl (mL/min)	Induction Dose of ValcYTE Tablets	Maintenance/Prevention Dose of ValcYTE Tablets
≥ 60	900 mg twice daily	900 mg once daily
40 – 59	450 mg twice daily	450 mg once daily
25 – 39	450 mg once daily	450 mg every 2 days
10 – 24	450 mg every 2 days	450 mg twice weekly
< 10	Not recommended	Not recommended

Creatinine clearance is calculated from serum creatinine by the following formulae:

For males:

$$\frac{(140 - \text{age}[\text{years}]) \times (\text{body weight} [\text{kg}])}{(72) \times (0.011 \times \text{serum creatinine} [\text{micromol/L}])}$$

For females:

$$0.85 \times \text{male value}$$

Children

Safety and efficacy have not been established in this patient population. The use of ValcYTE in children is not recommended because the pharmacokinetic characteristics of ValcYTE have not been established in this patient population (see section 3.2.5 *Pharmacokinetics in Special Populations*).

2.3 Contraindications

ValcYTE is contraindicated in patients with known hypersensitivity to valganciclovir, ganciclovir or to any of the excipients.

2.4 Warnings and Precautions

2.4.1 General

Cross hypersensitivity

Due to the similarity of the chemical structure of ganciclovir and that of aciclovir and penciclovir, a cross-hypersensitivity reaction between these drugs is possible. Caution should therefore be used when prescribing ValcYTE to patients with known hypersensitivity to aciclovir or penciclovir, (or to their prodrugs, valaciclovir or famciclovir respectively)

Mutagenicity, teratogenicity, carcinogenicity, fertility and contraception

In animal studies ganciclovir was found to be mutagenic, teratogenic, carcinogenic. ValcYTE should therefore be considered a potential teratogen and carcinogen in humans with the potential to cause birth defects and cancers. Prior to initiation of valganciclovir treatment, patients should be advised of the potential risks to the fetus and to use contraceptive measures. Based on clinical and nonclinical studies, ValcYTE may cause temporary or permanent inhibition of spermatogenesis (see Sections 2.5.1 Females and Males of Reproductive Potential, 2.5.2 Pregnancy, 2.5.3 Lactation, 2.6, *Undesirable effects*, 3.3 Nonclinical Safety and 4.2 *Special Instructions for Use, Handling and Disposal*).

Myelosuppression

ValcYTE should be used with caution in patients with pre-existing hematological cytopenia or a history of drug-related hematological cytopenia and in patients receiving radiotherapy.

Severe leucopenia, neutropenia, anemia, thrombocytopenia, pancytopenia, bone marrow depression and aplastic anemia have been observed in patients treated with ValcYTE (and ganciclovir). Therapy should not be initiated if the absolute neutrophil count is less than 500 cells/ μL or the platelet count is less than 25000 / μL or the hemoglobin is less than 8 g/dl (see Sections 2.2.1, *Special Dosage Instructions*, 2.4, *Warnings and Precautions* and 2.6, *Undesirable effects*).

It is recommended that complete blood counts and platelet counts be monitored in all patients during therapy, particularly in patients with renal impairment (see Sections 2.2.4 *Laboratory Test*).

In patients with severe leukopenia, neutropenia, anemia and/or thrombocytopenia treatment with hematopoietic growth factors and/or the interruption of therapy is recommended (see Sections 2.6 *Undesirable Effects*). Seizures have been reported in patients taking imipenem-cilastatin and ganciclovir. ValcYTE should not be used concomitantly with imipenem-cilastatin unless the potential benefits outweigh the potential risks (see Section 2.8, *Interactions with other Medicinal Products and other Forms of Interactions*).

Zidovudine and ValcYTE each have the potential to cause neutropenia and anemia. Some patients may not tolerate concomitant therapy at full dosage (see Section 2.8, *Interactions with other Medicinal Products and other Forms of Interactions*).

Didanosine plasma concentrations may increase during concomitant use with ValcYTE; therefore patients should be closely monitored for didanosine toxicity (see Section 28, *Interactions with other Medicinal Products and other Forms of Interactions*).

Concomitant use of other drugs that are known to be myelosuppressive or associated with renal impairment with ValcYTE may result in added toxicity (see Section 2.8 *Interactions with other Medicinal Products and other Forms of Interactions*). The bioavailability of ganciclovir from ValcYTE is 10-fold higher than from ganciclovir capsules. ValcYTE tablets cannot be substituted for ganciclovir capsules on a one-to-one basis. Patients switching from ganciclovir capsules should be advised of the risk of overdose if they take more than the prescribed number of ValcYTE tablets (see Section 2.2, *Dosage and Method of Administration* and 2.7 *Overdose*).

2.4.2 Drug Abuse and Dependence

No information is available for drug abuse and dependence with ValcYTE.

2.4.3 Ability to Drive and Use Machines

Adverse reactions such as seizures, dizziness, and confusion have been reported with the use of ValcYTE and/or ganciclovir (see Section 2.6 *Undesirable Effects*). If they occur, such effects may affect tasks requiring alertness including the patient’s ability to drive and operate machinery.

2.5 Use in Special Populations

2.5.1 Females and Males of Reproductive Potential

Fertility

In animal studies ganciclovir was found to impair fertility (see section 3.3.3 *Impairment of Fertility*). In a clinical study, renal transplant patients receiving ValcYTE for CMV prophylaxis for up to 200 days were compared to an untreated control group. Spermatogenesis was inhibited during treatment with ValcYTE. At follow-up, approximately six months after treatment discontinuation, the mean sperm density in treated patients was comparable to that observed in the untreated control group. In ValcYTE treated patients, all patients with normal sperm density (n=7) and 8/13 patients with low sperm density at baseline, had normal density after treatment cessation. In the control group, all patients with normal sperm density (n=6) and 2/4 patients with low sperm density at baseline, had normal density at the end of follow-up.

Contraception

Women of reproductive potential should be advised to use effective contraception during and for at least 30 days after treatment. Sexually active men are recommended to use condoms during and for at least 90 days after cessation of treatment with ValcYTE, unless it is certain that the female partner is not at risk of becoming pregnant (see Sections 2.4 *Warnings and Precautions* and 3.3.4 *Reproductive Toxicity*).

Women of childbearing potential should be advised to use effective contraception during treatment. Male patients should be advised to practice barrier contraception during and for at least 90 days following treatment with ValcYTE (see Sections 2.4, *Warnings and Precautions* and 3.3.4, *Teratogenicity*).

The safety of ValcYTE for use in human pregnancy has not been established. The use of ValcYTE should be avoided in pregnant women unless the benefit to the mother outweighs the potential risk to the fetus.

2.5.2 Pregnancy

The safety of ValcYTE for use in pregnant women has not been established. However, ganciclovir readily diffuses across the human placenta. The use of ValcYTE should be avoided in pregnant women unless the benefit to the mother outweighs the potential risk to the fetus. Reprotoxicity studies have not been repeated with valganciclovir because of the rapid and extensive conversion to ganciclovir. In animal studies ganciclovir was associated with reproductive toxicity and teratogenicity (see Sections 3.3.4 *Reproductive Toxicity*).

The safe use of ValcYTE during labor and delivery has not been established.

2.5.3 Lactation

Peri- and postnatal development has not been studied with valganciclovir or with ganciclovir but the possibility of ganciclovir being excreted in the breast milk and causing serious adverse reactions in the nursing infant cannot be discounted. Human data are not available but animal data indicates that ganciclovir is excreted in the milk of lactating rats. Therefore, a decision should be made to discontinue the drug or discontinue nursing taking into consideration the potential benefit of ValcYTE to the nursing mother.

2.5.4 Pediatric Use

The safety and efficacy of ValcYTE in pediatric patients have not been established in adequate and well-controlled clinical studies (see Section 2.2.1 *Special Dosage Instructions*).

2.5.5 Geriatric Use

Safety and efficacy have not been established in this patient population (see Section 2.2.1 *Special Dosage Instructions*).

2.5.6 Renal Impairment

In patients with impaired renal function, dosage adjustments based on creatinine clearance are required (see Sections 2.2.1, *Special Dosage Instructions* and 3.2.5, *Pharmacokinetics in Special Populations*).

2.5.7 Hepatic Impairment

Safety and efficacy have not been established in this patient population (see Section 2.2.1 *Special Dosage Instructions* and 3.2.5 *Pharmacokinetics in Special Populations*).

2.6 Undesirable Effects

2.6.1 Clinical Trials

Experience with ValcYTE

Valganciclovir is a prodrug of ganciclovir, which is rapidly converted to ganciclovir after oral administration. The undesirable effects known to be associated with ganciclovir usage can therefore be expected to occur with ValcYTE. All of the adverse events observed in ValcYTE clinical studies have been previously observed with ganciclovir. Therefore, adverse drug reactions reported with IV or oral ganciclovir (no longer available) or with valganciclovir are included in the table of adverse reactions (see Table 3). In patients treated with valganciclovir/ganciclovir the most serious and frequent adverse drug reactions are haematological reactions and include neutropenia, anemia and thrombocytopenia. The frequencies presented in the table of adverse reactions are derived from a pooled population of patients (n=1704) receiving maintenance therapy with ganciclovir (GAN 1697, GAN 1653, GAN 2304, GAN 1774, GAN 2226, AVI 034, GAN 041) or valganciclovir (WV15376, WV15705). Exception is made for anaphylactic reaction, agranulocytosis and granulocytopenia the frequencies of which are derived from post-marketing experience. Frequencies are presented as percentages and as CIOMS frequency categories defined as very common ($\geq 1/10$), common ($\geq 1/100$ to $< 1/10$), uncommon ($\geq 1/1,000$ to $< 1/100$), rare ($\geq 1/10,000$ to $< 1/1,000$) and very rare ($< 1/10,000$).

The overall safety profile of ganciclovir/valganciclovir is consistent in HIV and transplant populations except that retinal detachment has only been reported in patients with CMV retinitis. However, there are some differences in the frequency of certain reactions. Valganciclovir is associated with a higher risk of diarrhoea compared to intravenous ganciclovir. Pyrexia, candida infections, depression, severe neutropenia (ANC $< 500/\mu\text{L}$) and skin reactions are reported more frequently in patients with HIV. Renal and hepatic dysfunction are reported more frequently in organ transplant recipients.

Table 3 Frequency of Ganciclovir/Valganciclovir ADRs Reported in HIV Patients Receiving Maintenance Therapy (n=1704)

ADR (MedDRA) System Organ Class	Percentage	Frequency Category
Infections and infestations:		
Candida infections including oral candidiasis.	22.42%	Very common
Upper respiratory tract infection	16.26%	
Sepsis	6.92%	Common
Influenza	3.23%	
Urinary tract infection	2.35%	
Cellulitis	1.47%	
Blood and lymphatic disorders:		
Neutropenia	26.12%	Very common
Anemia	19.89%	
Thrombocytopenia	7.34%	Common
Leukopenia	3.93%	
Pancytopenia	1.06%	
Bone marrow failure	0.29%	
Aplastic anaemia	0.06%	Rare
Agranulocytosis*	0.02%	
Granulocytopenia*	0.02%	
Immune system disorders:		
Hypersensitivity	1.12%	Common
Anaphylactic reaction*	0.02%	Rare
Metabolic and nutrition disorders:		
Decreased appetite	12.09%	Very common
Weight decreased	6.46%	Common
Psychiatric disorders:		
Depression	6.69%	Common
Confusional state	2.99%	
Anxiety	2.64%	Uncommon
Agitation	0.59%	
Psychotic disorder	0.23%	
Thinking abnormal	0.18%	
Hallucinations	0.18%	
Nervous system disorders:		
Headache	17.37%	Very common
Insomnia	7.22%	Common
Neuropathy peripheral	6.16%	
Dizziness	5.52%	
Paraesthesia	3.58%	
Hypoaesthesia	2.58%	
Seizures	2.29%	
Dysgeusia (taste disturbance)	1.35%	Uncommon
Tremor	0.88%	

(DR (MedDRA) System Organ Class	Percentage	Frequency Category
Eye disorders:		
Visual impairment	7.10%	Common
Retinal detachment**	5.93%	
Vitreous floaters	3.99%	
Eye pain	2.99%	
Conjunctivitis	1.58%	
Macular oedema	1.06%	
Ear and labyrinth disorders:		
Ear pain	1.17%	Common
Deafness	0.65%	Uncommon
Cardiac disorders:		
Cardiac Aarrhythmias	0.47%	Uncommon
Vascular disorders:		
Hypotension	2.05%	Common
Respiratory, thoracic and mediastinal disorders:		
Cough	18.31%	Very common
Dyspnoea	11.80%	
Gastrointestinal disorders:		
Diarrhea	34.27%	Very common
Nausea	26.35%	
Vomiting	14.85%	Common
Abdominal pain	10.97%	
Dyspepsia	4.81%	
Flatulence	4.58%	
Abdominal pain upper	4.58%	
Constipation	3.70%	
Mouth ulceration	3.17%	
Dysphagia	2.93%	
Abdominal distention	2.41%	
Pancreatitis	1.64%	
Hepato-biliary disorders:		
Blood alkaline phosphatase increased	3.58%	Common
Hepatic function abnormal	3.23%	
Aspartate aminotransferase increased	1.88%	
Alanine aminotransferase increased	1.23%	
Skin and subcutaneous tissues disorders:		
Dermatitis	11.80%	Very common
Night sweats	7.92%	Common
Pruritus	4.58%	Uncommon
Rash	2.52%	
Alopecia	1.29%	
Dry skin	0.94%	
Urticaria	0.70%	
Musculo-skeletal and connective tissue disorders:		
Back pain	4.46%	Common
Myalgia	3.52%	
Arthralgia	3.35%	
Muscle spasms	2.99%	
Renal and urinary disorders:		
Renal impairment	2.52%	Common
Creatinine clearance renal decreased	2.35%	Uncommon
Blood creatinine increased	1.88%	
Renal failure	0.76%	
Hematuria	0.70%	
Reproductive system and breast disorders:		
Infertility male	0.23%	Uncommon
General disorders and administration site conditions:		
Pyrexia	33.51%	Very common
Fatigue	18.96%	Common
Pain	5.81%	
Chills	5.40%	
Malaise	2.11%	
Asthenia	2.00%	
Chest pain	0.88%	
* The frequencies of these adverse reactions are derived from post-marketing experience.		
** Retinal detachment has only been reported in HIV patients treated for CMV retinitis.		
Description of selected adverse reactions		
Neutropenia		
The risk of neutropenia is not predictable on the basis of the number of neutrophils before treatment. Neutropenia usually occurs during the first or second week of induction therapy. The cell count usually normalizes within 2 to 5 days after discontinuation of the drug or dose reduction (see Section 2.4, Warnings and Precautions).		
Thrombocytopenia		
Patients with low baseline platelet counts (< 100,000 / µl) have an increased risk of developing thrombocytopenia. Patients with iatrogenic immunosuppression due to treatment with immunosuppressive drugs are at greater risk of thrombocytopenia than patients with HIV (see Section 2.4, Warnings and Precautions).		
Severe thrombocytopenia may be associated with potentially life-threatening bleeding.		
Influence of treatment duration or indication on adverse reactions:		
Severe neutropenia (ANC <500/µL) is seen more frequently in CMV retinitis patients (16%) undergoing treatment with valganciclovir than in solid organ transplant patients receiving valganciclovir or oral ganciclovir. In patients receiving valganciclovir or oral ganciclovir until Day 100 post-transplant, the incidence of severe neutropenia was 5% and 3% respectively, whilst in patients receiving valganciclovir until Day 200 post-transplant the incidence of severe neutropenia was 10%.		
There was a greater increase in serum creatinine seen in solid organ transplant patients treated until Day 100 or Day 200 post-transplant with both valganciclovir and oral ganciclovir compared to CMV retinitis patients. However, impaired renal function is a feature more frequent in solid organ transplantation patients. The overall safety profile of Valcyte did not change with the extension of prophylaxis up to 200 days in		

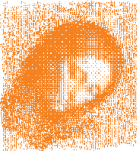
high risk kidney transplant patients. Leukopenia was reported with a slightly higher incidence in the 200 days arm while the incidence of neutropenia, anaemia and thrombocytopenia were similar in both arms.

Laboratory Abnormalities

Laboratory abnormalities reported in adult CMV retinitis patients and SOT patients receiving valganciclovir until Day 100 post-transplant are listed in Table 4. The incidence of laboratory abnormalities was comparable with the extension of prophylaxis up to 200 days in high risk kidney transplant patients.

Table 4: Laboratory Abnormalities

Laboratory abnormalities	CMV Retinitis Patients	Solid Organ Transplant Patients (Dosing until Day 100 Post-Transplant)	
	Valganciclovir (n=370)	Valganciclovir (n=244)	Oral ganciclovir (n=126)
	%	%	%
Neutropenia (ANC/ μ l)			
<500	16	5	3
500 - <750	17	3	2
750 - <1000	17	5	2
Anemia (hemoglobin g/dl)			
<6.5	7	1	2
6.5 - <8.0	10	5	7
8.0 - <9.5	14	31	25
Thrombocytopenia (platelets/ μ l)			
<25000	3	0	2
25000 - <50000	5	1	3
50000 - <100000	21	18	21
Serum creatinine (mg/dl)			
>2.5	2	14	21
>1.5 - 2.5	11	45	47



Didanosine

Didanosine plasma concentrations were found to be consistently raised when given with IV ganciclovir. At intravenous doses of 5 and 10 mg/kg/day, an increase in the AUC of didanosine ranging from 38 to 67% has been observed confirming a pharmacokinetic interaction during the concomitant administration of these drugs. There was no significant effect on ganciclovir concentrations. Patients should be closely monitored for didanosine toxicity (e.g. pancreatitis) (see Section 2.4 *Warnings and Precautions*).

Probenecid

Probenecid given with oral ganciclovir resulted in statistically significantly decreased renal clearance of ganciclovir (20%) leading to statistically significantly increased exposure (40%). These changes were consistent with a mechanism of interaction involving competition for renal tubular excretion. Therefore patients taking probenecid and valganciclovir should be closely monitored for ganciclovir toxicity.

3. PHARMACOLOGICAL PROPERTIES AND EFFECTS

3.1 Pharmacodynamic Properties

3.1.1 Mechanism of action

Valganciclovir is a L-valyl ester (prodrug) of ganciclovir, which after oral administration is rapidly converted to ganciclovir by intestinal and hepatic esterases. Ganciclovir is a synthetic analogue of 2'-deoxyguanosine, which inhibits replication of herpes viruses *in vitro* and *in vivo*. Sensitive human viruses include human cytomegalovirus (HCMV), herpes simplex virus-1 and -2 (HSV-1 and HSV-2), human herpes virus-6, 7 and 8 (HHV-6, HHV-7, HHV-8), Epstein-Barr virus (EBV), varicella-zoster virus (VZV) and hepatitis B virus.

In CMV-infected cells ganciclovir is initially phosphorylated to ganciclovir monophosphate by the viral protein kinase, UL97. Further phosphorylation occurs by cellular kinases to produce ganciclovir triphosphate, which is then slowly metabolised intracellularly. This has been shown to occur in HSV- and HCMV- infected cells with half-lives of 18 and between 6 and 24 hours respectively after removal of extracellular ganciclovir. As the phosphorylation is largely dependent on the viral kinase, phosphorylation of ganciclovir occurs preferentially in virus-infected cells.

The virustatic activity of ganciclovir is due to inhibition of viral DNA synthesis by: (a) competitive inhibition of incorporation of deoxyguanosine-triphosphate into DNA by viral DNA polymerase, and (b) incorporation of ganciclovir triphosphate into viral DNA causing termination of, or very limited further viral DNA elongation. Typical anti-viral IC₅₀ against CMV *in vitro* is in the range 0.08 µM (0.02 µg/ml) to 14 µM (3.5 µg/ml).

The clinical antiviral effect of Valcyte has been demonstrated in the treatment of AIDS patients with newly diagnosed CMV retinitis (clinical trial W15376). CMV shedding was decreased from 46% (32/69) of patients at study entry to 7% (4/55) of patients following four weeks of Valcyte treatment.

3.1.2 Clinical/ Efficacy Studies

Treatment of CMV retinitis

Clinical studies of Valcyte have been conducted in patients with AIDS and CMV retinitis. Valcyte has shown comparable efficacy for induction treatment of CMV retinitis to intravenous ganciclovir. Patients with newly diagnosed CMV retinitis were randomised in one study to induction therapy with either Valcyte or intravenous ganciclovir. The proportion of patients with progression of CMV retinitis at week 4 was the same in both treatment groups.

Following induction treatment dosing, patients in this study received maintenance treatment with Valcyte given at the dose of 900 mg daily. The mean (median) time from randomization to progression of CMV retinitis in the group receiving induction and maintenance treatment with Valcyte was 226 (160) days and in the group receiving induction treatment with intravenous ganciclovir and maintenance treatment with Valcyte was 219 (125) days.

Valcyte allows systemic exposure of ganciclovir similar to that achieved with recommended doses of intravenous ganciclovir, which has been shown to be efficacious in the treatment of CMV retinitis. Ganciclovir AUC has been shown to correlate with time to progression of CMV retinitis.

Prevention of CMV disease in transplantation

A double-blind, double-dummy clinical active comparator study has been conducted in heart, liver and kidney transplant patients at high-risk of CMV disease (D+/-R-) who received either Valcyte (900 mg od) or oral ganciclovir (1000 mg tid) starting within 10 days of transplantation until Day 100 post-transplant. The incidence of CMV disease (CMV syndrome + tissue invasive disease), as adjudicated by an independent Endpoint Committee, during the first 6 months post-transplant was 12.1% in the Valcyte arm (n=239) compared with 15.2% in the oral ganciclovir arm (n=125). The majority of cases occurred following cessation of prophylaxis (post Day 100) with cases in the valganciclovir arm occurring on average later than those in the oral ganciclovir arm. The incidence of acute rejection in the first 6 months was 29.7% in patients randomised to valganciclovir compared with 36.0% in the oral ganciclovir arm.

A double-blind, placebo controlled study has been conducted in 326 kidney transplant patients at high risk of CMV disease (D+/-R-) to assess the efficacy and safety of extending Valcyte CMV prophylaxis from 100 to 200 days post-transplant. Patients were randomised (1:1) to receive Valcyte tablets (900 mg od) within 10 days of transplantation either until Day 200 post-transplant or until Day 100 post-transplant followed by 100 days of placebo. Extending CMV prophylactic therapy with Valcyte until Day 200 post-transplant demonstrated superiority in preventing CMV disease within the first 12 months post-transplant in high risk kidney transplant patients compared to the 100 day dosing regimen.

The proportion of patients who developed CMV disease during the first 12 months post-transplant is shown in Table 5.

Table 5 Percentage of Kidney Transplant Patients with CMV Disease ¹ , 12 Month ITT Population			
	Valganciclovir 900 mg od 100 days	Valganciclovir 900 mg od 200 days	Cochran-Mantel-Haenszel p-value
Patients with confirmed or assumed CMV disease ²	71/163 (43.6%)	36/155 (23.2%)	0.0001
Patients with confirmed CMV disease	60/163 (36.8%)	25/155 (16.1%)	<0.0001

¹ CMV Disease is defined as either CMV syndrome or tissue invasive CMV.

² Confirmed CMV is a clinically confirmed case of CMV disease. Patients were assumed to have CMV disease if there was either no week 52 assessment or no confirmation of CMV disease before this time point.

The graft survival rate at 12 months post-transplant was 98.1% (160/163) for the 100 day dosing regimen and 98.2% (152/155) for the 200 day dosing regimen. The incidence of biopsy proven acute rejection at 12 months post-transplant was 17.2% (28/163) for the 100 day dosing regimen and 11.0% (17/155) for the 200 day dosing regimen.

Viral Resistance

Virus resistant to ganciclovir can arise after chronic dosing with valganciclovir by selection of mutations in either the viral kinase gene (UL97) responsible for ganciclovir monophosphorylation or the viral polymerase gene (UL54). UL97 mutations arise earlier and more frequently than mutations in UL54. Virus containing mutations in the UL97 gene is resistant to ganciclovir alone, with M460V/I, H520Q, C592G, A594V, L595S, C603W being the most frequently reported ganciclovir resistance-associated substitutions. Mutations in the UL54 gene may show cross-resistance to other antivirals targeting the viral polymerase, and vice versa. Amino acid substitutions in UL54 conferring cross-resistance to ganciclovir and cidofovir are generally located within the exonuclease domains and region V, however amino acid substitutions conferring cross-resistance to foscarnet are diverse, but concentrate at and between regions II (codon 696-742) and III (codon 805-845).

Treatment of CMV retinitis (Adult patients):

Genotypic analysis of CMV in polymorphonuclear leukocytes (PMNL) isolates from 148 patients with CMV retinitis enrolled in one clinical study has shown that 2.2%, 6.5%, 12.8%, and 15.3% contain UL97 mutations after 3, 6, 12 and 18 months, respectively, of valganciclovir treatment.

Prevention of CMV disease in transplantation:

Resistance was studied by genotypic analysis of CMV in PMNL samples collected i) on Day 100 (end of study drug prophylaxis) and ii) in cases of suspected CMV disease up to 6 months after transplantation. From the 245 patients randomised to receive valganciclovir, 198 Day 100 samples were available for testing and no ganciclovir resistance mutations were observed. This compares with 2 ganciclovir resistance mutations detected in the 103 samples tested (1.9%) for patients in the oral ganciclovir comparator arm.

Of the 245 patients randomised to receive valganciclovir, samples from 50 patients with suspected CMV disease were tested and no resistance mutations were observed. Of the 125 patients on the ganciclovir comparator arm, samples from 29 patients with suspected CMV disease were tested, from which two resistance mutations were observed, giving an incidence of resistance of 6.9%.

Resistance was evaluated in a study that extended valganciclovir CMV prophylaxis from 100 days to 200 days post-transplant in adult kidney transplant patients at high risk for CMV disease (D+/-R-) (see section 3.1.2 *Clinical/Efficacy Studies*). Five subjects from the 100 day group and four subjects from the 200 day group meeting the resistance analysis criteria had known ganciclovir resistance-associated amino acid substitutions detected. In six subjects, the following resistance associated amino acid substitutions were detected within pUL97: 100 day group: A440V, M460V, C592G; 200 day group: M460V, C603W. In three subjects, the following resistance-associated amino acid substitutions were detected within pUL54: 100 day group: E315D, 200 day group: E315D, P522S. Overall, the detection of known ganciclovir resistance-associated amino acid substitutions was observed more frequently in patients during prophylaxis therapy than after the completion of prophylaxis therapy (during therapy: 5/12 [42%] versus after therapy: 4/58 [7%]). The possibility of viral resistance should be considered in patients who show poor clinical response or experience persistent viral excretion during therapy.

3.1.3 Immunogenicity

Not applicable.

3.2 Pharmacokinetic Properties

The pharmacokinetic properties of valganciclovir have been evaluated in HIV- and CMV-seropositive patients, patients with AIDS and CMV retinitis and in solid organ transplant patients.

The parameters which control the exposure of ganciclovir from valganciclovir are bioavailability and renal function. The bioavailability of ganciclovir from valganciclovir is comparable across all the patient populations studied. The systemic exposure of ganciclovir to heart, kidney and liver transplant recipients was similar after oral administration of valganciclovir according to the renal function dosing algorithm.

3.2.1 Absorption

Valganciclovir is a prodrug of ganciclovir, which is well absorbed from the gastrointestinal tract and rapidly metabolised in the intestinal wall and liver to ganciclovir. The bioavailability of ganciclovir from oral dosing of valganciclovir is approximately 60%. Systemic exposure to valganciclovir is transient and low, AUC_{0-24h} and C_{max} values are approximately 1% and 3% of those of ganciclovir, respectively.

When valganciclovir was given with food at the recommended dose of 900 mg, increases were seen in both mean ganciclovir AUC_{0-24h} (approximately 30%) and mean ganciclovir C_{max} values (approximately 14%). Therefore, it is recommended that Valcyte be administered with food (see Section 2.2 *Dosage and Administration*).

3.2.2 Distribution

Because of rapid conversion of valganciclovir to ganciclovir, protein binding of Valcyte was not determined. Plasma protein binding of ganciclovir was 1-2% over concentrations of 0.5 and 51 µg/ml. The steady state volume of distribution of ganciclovir after intravenous administration was 0.680 ± 0.161 L/kg.

3.2.3 Metabolism

Valganciclovir is rapidly hydrolysed to ganciclovir; no other metabolites have been detected. Ganciclovir itself is not metabolized to a significant extent.

3.2.4 Elimination

Following dosing with Valcyte, renal excretion was ganciclovir by glomerular filtration and active tubular secretion is the major route of elimination of Valcyte. Renal clearance accounts for 81.5% ±22% of the systemic clearance of ganciclovir.

3.2.5 Pharmacokinetics in Special Populations

Geriatric Population

No investigations on valganciclovir or ganciclovir pharmacokinetics in adults older than 65 years of age have been undertaken. However as valganciclovir is a pro-drug of ganciclovir and because ganciclovir is mainly renally excreted and since renal clearance decreases with age, a decrease in ganciclovir total body clearance and a prolongation of ganciclovir half-life can be anticipated in elderly (see Section 2.2.1 *Special Dosage Instructions*).

Patients with renal impairment

The pharmacokinetics of ganciclovir from a single oral dose of 900 mg valganciclovir were evaluated in 24 otherwise healthy individuals with renal impairment.

Table 6: Pharmacokinetic parameters of ganciclovir from a single oral dose of 900 mg Valcyte tablets in patients with various degrees of renal impairment

Estimated Creatinine Clearance (mL/min)	N	Apparent Clearance (mL/min) Mean ± SD	AUC _{last} (µg·h/mL) Mean ± SD	Half-life (hours) Mean ± SD
51-70	6	249 ± 99	50.5 ± 23	4.9 ± 1.4
21-50	6	136 ± 64	100 ± 54	10.2 ± 4.4
11-20	6	45 ± 11	252 ± 64	21.8 ± 5.2
≤10	6	12.8 ± 8	407 ± 83	68.1 ± 35

Decreasing renal function resulted in decreased clearance of ganciclovir from valganciclovir with a corresponding increase in terminal half-life. Therefore, dosage adjustment is required for renally impaired patients (see Sections 2.2.1 Special Dosage Instructions and 2.4 Warnings and Precautions).

Patients undergoing hemodialysis

Ganciclovir is readily removable by hemodialysis. Data obtained during intermittent haemodialysis in patients dosed with valganciclovir showed estimated dialysis clearance as 138 mL/min ± 9.1% (N = 3) and intra-dialysis half-life estimated to 3.47 h (N = 6). 55% of ganciclovir was removed during a 3 hour dialysis session.

Stable liver transplant patients

The pharmacokinetics of ganciclovir from valganciclovir in stable liver transplant patients were investigated in one open label 4-part crossover study (N=28). The bioavailability of ganciclovir from valganciclovir, following a single dose of 900 mg valganciclovir under fed conditions, was approximately 60%. Ganciclovir AUC_{0-24h} was comparable to that achieved by 5 mg/kg intravenous ganciclovir in liver transplant patients.

Hepatic impairment

No pharmacokinetic study has been conducted and no population PK data was collected in patients with hepatic impairment undergoing valganciclovir therapy.

Patients with cystic fibrosis

In a phase I pharmacokinetic study, steady state systemic exposure to ganciclovir was assessed in lung transplant recipients with or without cystic fibrosis (N=31) who were receiving 900 mg/day of Valcyte as part of their post-transplant prophylaxis. The study indicated that cystic fibrosis had no statistically significant influence on the overall average systemic exposure to ganciclovir in lung transplant recipients. Ganciclovir exposure in lung transplant recipients was comparable to that shown to be efficacious in the prevention of CMV disease in other solid organ transplant recipients.

3.3 Nonclinical Safety

3.3.1 Carcinogenicity

Valganciclovir and ganciclovir were mutagenic in mouse lymphoma cells and clastogenic in mammalian cells. Such results are consistent with the positive mouse carcinogenicity study with ganciclovir. Ganciclovir is a potential carcinogen.

3.3.2 Genotoxicity

Valganciclovir and ganciclovir were mutagenic in mouse lymphoma cells and clastogenic in mammalian cells.

3.3.3 Impairment of Fertility

Ganciclovir causes impaired fertility and teratogenicity in animals. Reprotoxicity studies have not been repeated with valganciclovir because of the rapid and extensive conversion to ganciclovir. The same reprotoxicity warning is seen as applying to both drugs (see Section 2.4 *Warnings and Precautions*).

Based upon animal studies where asperma was induced at ganciclovir systemic exposures below therapeutic levels, it is considered likely that ganciclovir (and valganciclovir) could cause inhibition of human spermatogenesis (see Section 2.5.1 *Females and Males of Reproductive Potential, Fertility*).

3.3.4 Reproductive Toxicity

Ganciclovir causes teratogenicity in animals. Reprotoxicity studies have not been repeated with valganciclovir because of the rapid and extensive conversion to ganciclovir. The same reprotoxicity warning is seen as applying to both drugs (see Section 2.4 *Warnings and Precautions*).

4. PHARMACEUTICAL PARTICULARS

4.1 Storage

Film-coated Tablets:

Do not store above 30°C.

4.2 Special Instructions for Use, Handling and Disposal

Stability

Tablets should not be broken or crushed. Since Valcyte is considered a potential teratogen and carcinogen in humans, caution should be observed in handling broken tablets (see Warnings and Precautions). Avoid direct contact of broken or crushed tablets with skin or mucous membranes. If such contact occurs, wash thoroughly with soap and water, rinse eyes thoroughly with sterile water or plain water if sterile water is not available.

This medicine should not be used after the expiry date (EXP) shown on the pack.

Disposal of unused/ expired medicines

The release of pharmaceuticals in the environment should be minimized. Medicines should not be disposed of via wastewater and disposal through household waste should be avoided. Use established "collection systems" if available in your location.

4.3 Packs

Film-coated tablets 450 mg 60

Medicine: keep out of sight and reach of children

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